Spiral similarity and the configuration toolkit

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L5-GE2 Level 5 Advanced · Geometry

Spiral similarity centres, the Miquel point, symmedians, isogonal conjugates and the mixtilinear incircle, proved and used as tools.
By the end you can find the centre of a spiral similarity and the Miquel point of four lines; recognise symmedians and isogonal conjugates and prove their standard properties; use the incentre-excentre lemma, the shooting lemma and mixtilinear facts in full proofs.
Prerequisites: Olympiad angle chasing (L4-GE1), Lengths and ratios (L4-GE2). **Contests:** BMO2, EGMO, USAMO, IMO. **Time:** about 180 minutes.

1 Lesson

Let ABC be a triangle with $AB \neq AC$. Points D and E move on sides AB and AC so that $BD = CE$. Prove that the circle ADE passes through a fixed point other than A. Taking $D = B$, $E = C$ shows the fixed point must lie on the circumcircle Γ, and an accurate drawing suggests the midpoint N of arc BAC. Angle chasing at N does not start, because nothing yet relates N to D and E. What relates them is that N is the centre of a rotation taking D to E and B to C. This unit develops that view (spiral similarity), its main consequence (the Miquel point), and then four configurations that appear in BMO2 and IMO geometry: symmedians, isogonal conjugates, the incentre-excentre lemma and the mixtilinear incircle. Directed angles $\measuredangle$ modulo 180° are used as in L4-GE1.

1.1 1. Spiral similarity

A *spiral similarity* with centre S, angle θ and ratio $k > 0$ is a rotation about S through θ followed by the homothety at S with ratio k. In complex numbers it is $z \mapsto s + ke^{i\theta}(z - s)$.

Theorem 1.1.
If $A \neq B$, $C \neq D$ and $\overrightarrow{AB} \neq \overrightarrow{CD}$, there is exactly one spiral similarity sending A to C and B to D.

Theorem 1.2. In Theorem 1.1, suppose lines AC and BD meet at X. Then the centre S is the second intersection of circles XAB and XCD (or X itself, if these circles touch at X).

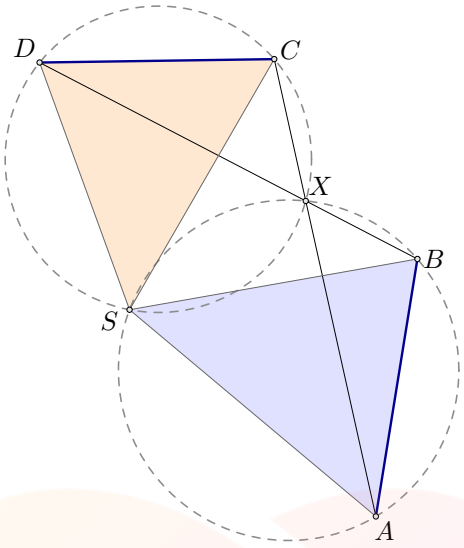
Theorem 1.3. The spiral similarity sending $A \mapsto C$, $B \mapsto D$ and the one sending $A \mapsto B$, $C \mapsto D$ have the same centre.

Proof. For 1.1, the map is $z \mapsto \alpha z + \beta$ with $\alpha = \frac{d-c}{b-a}$ and $\beta = c - \alpha a$; since $\alpha \neq 1$ it has the fixed point $s = \beta/(1-\alpha)$, and any map sending $a \mapsto c$, $b \mapsto d$ of this form is this one. For 1.3, if S sends $A \mapsto C$, $B \mapsto D$, then $SC/SA = SD/SB$ and $\measuredangle ASC = \measuredangle BSD$, so triangles SAB and SCD are directly similar and so are triangles SAC and SBD (SAS); the second similarity is a spiral similarity at S sending $A \mapsto B$, $C \mapsto D$. For 1.2, from $SAC \sim SBD$,

$$\measuredangle SAX = \measuredangle SAC = \measuredangle SBD = \measuredangle SBX,$$

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so S, A, B, X are concyclic, and similarly $\angle SCX = \angle SDX$ gives S, C, D, X concyclic. If $S = X$ then X is the centre and $AB \parallel CD$, so the circles are homothetic at X and touch there. ■



Solution of the opening problem. Lines DB and EC meet at A, so by Theorem 1.2 the centre M of the spiral similarity $D \mapsto B$, $E \mapsto C$ lies on circles ADE and ABC. By Theorem 1.3, M is also the centre of the similarity $D \mapsto E$, $B \mapsto C$, whose ratio is $EC/DB = 1$. So it is a rotation, and $MB = MC$. Its angle is the angle from $\overrightarrow{DB}$ to $\overrightarrow{EC}$, which is the oriented angle BAC, so M sees BC at the same oriented angle as A and $M = N$. (If $AB = AC$ then $DE \parallel BC$, circle ADE touches Γ at A, and $N = A$; this is why that case is excluded.)

Corollary 1.4.
Circles ω_1, ω_2 meet at P and Q. If a line through P meets ω_1 again at A and ω_2 again at B, then Q is the centre of a spiral similarity sending ω_1 to ω_2 and A to B, the same one for every such line.

Proof. Take two such lines, giving A, B and A_1, B_1. The lines AB and A_1B_1 joining corresponding points of the similarity $A \mapsto B$, $A_1 \mapsto B_1$ meet at P, so by Theorem 1.2 its centre is the second intersection of $\omega_1 = (PAA_1)$ and $\omega_2 = (PBB_1)$, namely Q (the circles meet twice, so they do not touch). This similarity fixes Q, so it sends circle $QAA_1 = \omega_1$ to circle $QBB_1 = \omega_2$. A spiral similarity is determined by its centre and the image of one point, so fixing the first line, every other line gives the same map. ■

Example 1.
Circles ω_1 and ω_2 meet at P and Q. A line through P meets ω_1 again at A and ω_2 again at B, and the tangents to ω_1 at A and to ω_2 at B meet at T. Prove that A, B, Q, T are concyclic.

Solution.
With directed angles the tangent-chord angle in ω_1 and in ω_2 gives $\angle(TA, AB) = \angle AQP$ and $\angle(AB, BT) = \angle PQB$, and adding gives $\angle ATB = \angle AQB$. Corollary 1.4 gives a second proof, which shows more: any line through A and its image under the similarity below meet on circle ABQ. The spiral similarity at Q sends ω_1 to ω_2 and A to B, so it sends the tangent at A to the tangent at B. A spiral similarity with angle θ turns every line through θ, so $\angle(TA, TB) = \theta = \angle(QA, QB)$. Hence $\angle ATB = \angle AQB$ and the four points are concyclic.

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(If the tangents are parallel, $\theta = 0$, which is impossible since Q is not on line AB.)

1.2 2. The Miquel point

Theorem 2.1 (Miquel).
Let $ABCD$ be a quadrilateral with no two sides parallel, $Q = AB \cap CD$ and $R = AD \cap BC$. Then circles QAD, QBC, RAB, RCD pass through one point M, the Miquel point. It is the centre of the spiral similarity $A \mapsto B$, $D \mapsto C$ and of the spiral similarity $A \mapsto D$, $B \mapsto C$.
Theorem 2.2. If moreover $ABCD$ is convex and cyclic with centre O, then M lies on QR, $OM \perp QR$, and $QR^2 = QA \cdot QB + RA \cdot RD$.

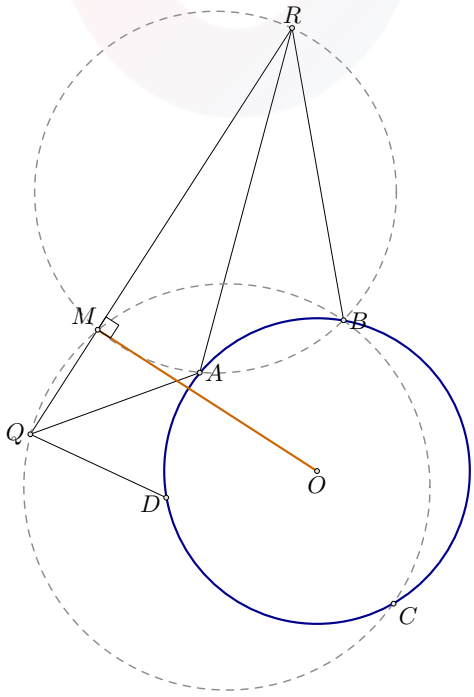
Proof. of 2.1. The map $A \mapsto B$, $D \mapsto C$ is not a translation, as $AB \nparallel DC$. Lines AB and DC meet at Q, so by Theorem 1.2 its centre lies on circles QAD and QBC. By Theorem 1.3 the same point is the centre of $A \mapsto D$, $B \mapsto C$, and lines AD, BC meet at R, so it lies on circles RAB and RDC. (The circles QAD, QBC do not touch at Q: then Q would be the centre and would lie on circle RAB, which meets line AB only at A and B.) ■

Proof. of 2.2. From circles $QBCM$ and $RDCM$,

$$\measuredangle QMR = \measuredangle QMC + \measuredangle CMR = \measuredangle QBC + \measuredangle CDR = \measuredangle ABC + \measuredangle CDA = 0,$$

as $ABCD$ is cyclic; so Q, M, R are collinear. Power of Q in circle $RABM$ and of R in circle $QADM$ give $QA \cdot QB = QM \cdot QR$ and $RA \cdot RD = RM \cdot RQ$ (signed along line QR). Adding gives $QR^2 = QA \cdot QB + RA \cdot RD$. Subtracting, and writing the powers with respect to the circumcircle as $OQ^2 - \rho^2$ and $OR^2 - \rho^2$, gives $OQ^2 - OR^2 = MQ^2 - MR^2$, so O lies on the perpendicular to QR through M. ■

Compare Example 4 of L5-GE1, where the same identity came from Brocard's theorem; in that language M is the inverse of $P = AC \cap BD$ in the circumcircle.



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Example 2.
Let $ABCD$ be a convex quadrilateral with $AD = BC$ and no two sides parallel, and let $Q = AB \cap CD$. Prove that the perpendicular bisectors of AB and CD meet on the circle QBC.

Solution.
Equal lengths $AD = BC$ on two sides suggest a rotation taking one to the other, and its centre would be equidistant from the corresponding endpoints. By Theorem 2.1 the Miquel point M is the centre of the spiral similarity $A \mapsto B$, $D \mapsto C$, whose ratio is $BC/AD = 1$. So it is a rotation, and $MA = MB$, $MD = MC$: M lies on both perpendicular bisectors. These bisectors are not parallel (as $AB \nparallel CD$), so they meet only at M, and M lies on circle QBC by Theorem 2.1.

1.3 3. Symmedians

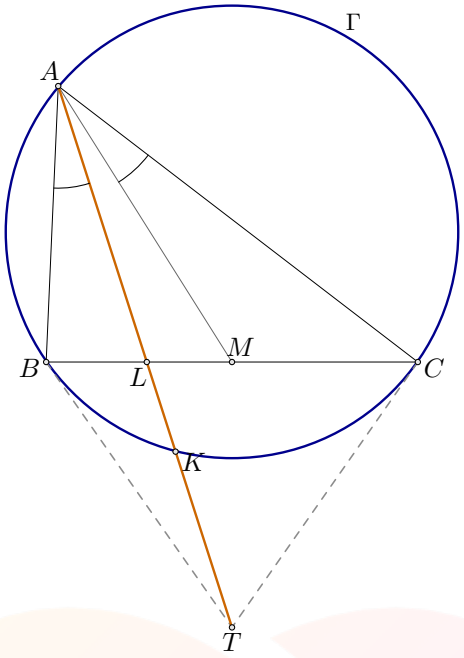
The *A-symmedian* is the reflection of the median AM in the bisector of angle A. Two lines through A symmetric in that bisector are called *isogonal*.

Theorem 3.1.
Let the A-symmedian meet BC at L and Γ again at K. (1) $BL/LC = AB^2/AC^2$. (2) $KB/KC = AB/AC$, and K is the only point $X \neq A$ of Γ with $XB/XC = AB/AC$. (3) If $\angle A \neq 90^\circ$, the tangents to Γ at B and C meet on line AK. (4) The symmedian bisects every segment XY with X on line AB, Y on line AC and B, C, Y, X concyclic (an antiparallel to BC).

(1) is the ratio lemma (L4-GE2): $\frac{BL}{LC} = \frac{AB \sin \angle BAL}{AC \sin \angle LAC}$, while for the median $1 = \frac{AB \sin \angle BAM}{AC \sin \angle MAC}$; isogonality swaps the two sines, so $\frac{BL}{LC} = \frac{AB}{AC} \cdot \frac{AB}{AC}$. For (2), $\sin \angle BAK = KB/2R$ and $\sin \angle KAC = KC/2R$, so (1) gives $KB/KC = AB/AC$. The points X with $XB/XC = AB/AC$ form a circle (or the perpendicular bisector of BC if $AB = AC$), which meets Γ in at most two points, A and K. For (3), let T be the intersection of the tangents and let line AT meet Γ again at K'. By power of a point $TB^2 = TK' \cdot TA$, so triangles TBK' and TAB are similar (common angle at T) and $BK'/AB = TB/TA$; likewise $CK'/AC = TC/TA$. As $TB = TC$, $K'B/K'C = AB/AC$, and $K' = K$ by (2). For (4), reflection in the bisector of angle A turns antiparallels into parallels to BC and the symmedian into the median, which bisects every segment parallel to BC. ■

Statement (3) is the form in which symmedians usually appear in problems: you are given the tangents at B and C, and you should recognise AT as the symmedian.

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Example 3.
In triangle ABC, the circle through B tangent to line AC at A and the circle through C tangent to line AB at A meet again at S. Let K be the reflection of A in S. Prove that K lies on Γ and that AK is the A-symmedian.

Solution.
The circles touch different lines at A, so they cross there and $S \neq A$ exists. The tangent conditions give $\measuredangle(AC, AS) = \measuredangle ABS$ and $\measuredangle(AB, AS) = \measuredangle ACS$, which rearrange to $\measuredangle SBA = \measuredangle SAC$ and $\measuredangle SAB = \measuredangle SCA$. So triangles SBA and SAC are directly similar, and S is the centre of the spiral similarity $B \mapsto A$, $A \mapsto C$. Its angle is $\theta = \measuredangle BSA = \measuredangle SBA + \measuredangle BAS = \measuredangle SAC + \measuredangle BAS = \measuredangle BAC$, and $SB/SA = SA/SC = AB/AC$.
Now $SK = SA = \sqrt{SB \cdot SC}$ and $\measuredangle BSK = \measuredangle BSA = \theta = \measuredangle ASC = \measuredangle KSC$, so triangles BSK and KSC are directly similar (SAS; as K is on the ray opposite to SA, the rotations from ray SB to ray SK and from ray SK to ray SC both equal the angle of the spiral similarity plus $180°$). Hence

$$\measuredangle BKC = \measuredangle BKS + \measuredangle SKC = \measuredangle BKS + \measuredangle SBK = \measuredangle BSK = \measuredangle BAC,$$

so K lies on Γ. The same similarity gives $KB/KC = SB/SK = SB/SA = AB/AC$, so AK is the symmedian by Theorem 3.1(2). In particular S is the midpoint of the symmedian chord.

1.4 4. Isogonal conjugates

Theorem 4.1.
Let P be a point inside triangle ABC. The reflections of AP, BP, CP in the bisectors of the angles at A, B, C meet at a point P^*, the isogonal conjugate of P. The six feet of the perpendiculars from P and P^* to the sides lie on one circle, centred at the midpoint of PP^*.

Proof. By trigonometric Ceva,

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$$\frac{\sin\angle BAP}{\sin\angle PAC}\cdot\frac{\sin\angle CBP}{\sin\angle PBA}\cdot\frac{\sin\angle ACP}{\sin\angle PCB}=1.$$

Reflecting the three lines swaps the two angles in each fraction, so the product for the reflected lines is the reciprocal, also 1, and they concur. For the circle, let F, E be the feet from P on AB, AC and F', E' those from P^*. Then $AF = AP\cos\angle PAB$, $AE' = AP^*\cos\angle P^*AC = AP^*\cos\angle PAB$, and similarly $AE = AP\cos\angle PAC$, $AF' = AP^*\cos\angle PAC$. So $AF\cdot AF' = AE\cdot AE'$ and E, E', F, F' are concyclic. The midpoint N of PP^* projects onto AB at the midpoint of FF', so N is on the perpendicular bisector of FF', and similarly of EE'; this circle has centre N. The same holds at B, and the two circles share the centre N and the point F, so they coincide. ■

The pairs to know are $I \leftrightarrow I$, $O \leftrightarrow H$ (since $\angle BAH = 90^\circ - \angle B = \angle OAC$), and the centroid with the symmedian point. For O and H the circle is the nine-point circle.

1.5 5. The incentre and the mixtilinear incircle

Write M_A, M_B, M_C for the midpoints of the arcs BC, CA, AB not containing the opposite vertex. Recall the incentre-excentre lemma (L4-GE1): $M_AB = M_AC = M_AI = M_AI_A$. The *A-mixtilinear incircle* ω_A is the circle tangent to AB, AC and internally tangent to Γ; let it touch AB, AC, Γ at K, L, T. Since ω_A lies inside angle BAC and inside Γ, T is on arc BC not containing A.

Theorem 5.1 (shooting lemma).
If a circle inside Γ touches Γ at T and a chord XY at Z, then line TZ passes through the midpoint W of arc XY not containing T, and $WZ\cdot WT = WX^2$.

Theorem 5.2. (1) I is the midpoint of KL. (2) B, K, I, T are concyclic, and so are C, L, I, T. (3) Line TI passes through the midpoint N of arc BAC.

Proof. of 5.1. The homothety at T taking the small circle to Γ takes line XY to the tangent of Γ parallel to XY on the other side of XY from T, and Z to its point of contact, which is W. Then $\angle WXZ = \angle WXY = \angle WTX$ (equal arcs WX, WY), so triangles WXZ and WTX are similar and $WX^2 = WZ\cdot WT$. ■

Proof. of 5.2. (1) By 5.1 with chord AB, T, K, M_C are collinear and $M_CK\cdot M_CT = M_CA^2 = M_CI^2$, the last by the incentre-excentre lemma at vertex C. So triangles M_CIK and M_CTI are similar, with opposite orientation, and $\measuredangle(IK, IM_C) = \measuredangle(TM_C, TI)$. In the same way $\measuredangle(IM_B, IL) = \measuredangle(TI, TM_B)$. Adding,

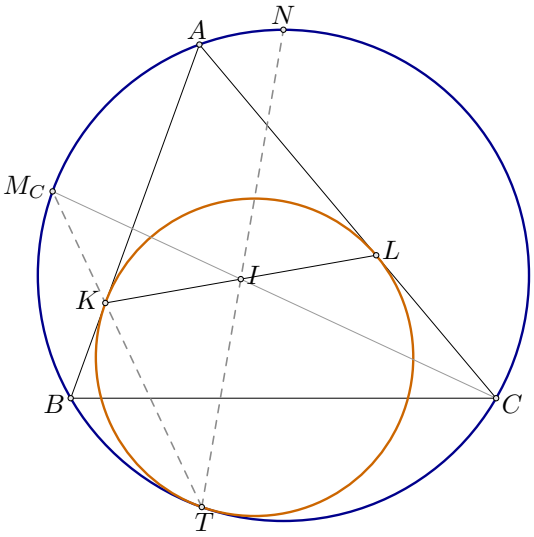
$$\measuredangle(IK, IL) = \measuredangle M_CTM_B + \measuredangle M_CIM_B = \measuredangle M_CAM_B + \measuredangle M_CIM_B.$$

Since $M_BA = M_BI$ and $M_CA = M_CI$, line M_BM_C is the perpendicular bisector of AI, so $\measuredangle M_CAM_B = -\measuredangle M_CIM_B$ and K, I, L are collinear. As $AK = AL$ and AI bisects angle KAL, I is the midpoint of KL and $KL \perp AI$.

(2) Since triangle AIK has a right angle at I, $AK = AI/\cos\frac{A}{2} = 2r/\sin A = bc/s$, which is less than c as b is less than s; so K lies on side AB. In triangle BKI, $\angle BKI = 180^\circ - \angle AKI = 90^\circ + \frac{A}{2}$ and $\angle KBI = \frac{B}{2}$, so $\angle KIB = \frac{C}{2}$. Also $\angle KTB = \angle M_CTB = \angle M_CCB = \frac{C}{2}$. As I and T lie on the same side of AB, B, K, I, T are concyclic; similarly for C, L, I, T.

(3) By (2), $\measuredangle BTI = \measuredangle BKI = \measuredangle(AB, KL)$ and $\measuredangle ITC = \measuredangle ILC = \measuredangle(KL, AC)$. These are equal, because reflection in AI swaps lines AB and AC and fixes line $KL \perp AI$. So TI bisects angle BTC. Triangle ABC lies inside angle BTC, so I does and TI is the internal bisector; it passes through the midpoint of arc BC not containing T, which is N. ■

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Common traps.

- Pairing points wrongly in Theorem 1.2: the circles contain the two segments being mapped to each other, and X is where the lines joining corresponding points meet.
- Forgetting the degenerate case of a spiral similarity: a translation has no centre.
- Calling AT a median, or using Theorem 3.1(3) when $\angle A = 90^\circ$ and the tangents are parallel.
- Using the shooting lemma with the wrong arc midpoint; it is the arc on the far side of the chord from T.
- Quoting mixtilinear facts without proof in a BMO2 script. Facts beyond the incentre-excentre lemma should be proved in a few lines.

1.6 How to spot this in a contest

1. Two circles through a common point and a line through it: the other intersection is a spiral centre (Corollary 1.4).
2. Four lines, or a quadrilateral with its opposite sides extended: find the Miquel point before anything else.
3. Equal lengths on two non-parallel segments: a rotation, whose centre is on perpendicular bisectors.
4. Tangents at B and C meeting at a point: the A-symmedian. Equal angles $\angle BAX = \angle YAC$: isogonal lines, and look for isogonal conjugates.
5. An incentre with an arc midpoint, or a circle tangent to two sides and to Γ: the incentre-excentre lemma and Theorems 5.1 and 5.2.

These configurations appear throughout *Beyond the Incenter*; the cross-ratio chapter there connects Section 3 with harmonic quadrilaterals ($ABKC$ above is one).

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2 Practice problems

The problems are roughly in order of difficulty, from ⋆⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆⋆ Triangle ABC has $BC = 8$, $CA = 9$ and $AB = 7$. The A-symmedian meets the circumcircle again at K. Find AK.

Problem 2 ⋆⋆ Circles ω_1 and ω_2, with centres O_1 and O_2, meet at P and Q. A line through P meets ω_1 again at A and ω_2 again at B. Prove that lines AO_1 and BO_2 meet at a point of the circle through Q, O_1 and O_2.

Problem 3 ⋆⋆⋆ Point P lies inside triangle ABC with $\angle PBA = \angle PCA$. Let D, E, F be the feet of the perpendiculars from P to BC, CA, AB, and let M be the midpoint of BC. Prove that D, E, F, M lie on a circle.

Problem 4 ⋆⋆⋆ Let ABC be an acute triangle with circumcircle Γ. The tangents to Γ at B and C meet at T. Points D on line AB and E on line AC are such that T is the midpoint of DE. Prove that B, C, E, D are concyclic.

Problem 5 ⋆⋆⋆ Let $ABCD$ be a convex quadrilateral in which no two sides are parallel. Lines AB and CD meet at Q, and lines AD and BC meet at R. Prove that the circumcircles of triangles QAD and RAB meet at a second point M, and that M lies on the perpendicular bisector of AC if and only if

$$AB \cdot AD = CB \cdot CD.$$

Problem 6 ⋆⋆⋆⋆ The A-mixtilinear incircle of triangle ABC touches the circumcircle at T, and the A-excircle touches BC at X. With $a = BC$, $b = CA$, $c = AB$ and s the semiperimeter, prove that

$$\frac{TB}{TC} = \frac{c(s-b)}{b(s-c)},$$

and deduce that $\angle BAT = \angle XAC$.

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **BMO2 2022/23 Problem 1** ⋆⋆
   The centres of circles ABI and ACI are arc midpoints of the circumcircle.
2. **IMO 2006 Problem 1** ⋆⋆
   Show that P lies on circle BIC and use the centre of that circle.
3. **BMO2 2012/13 Problem 2** ⋆⋆⋆
   The conclusion says AP and AQ are isogonal; transfer the equal angles through the parallelogram.

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4. **BMO2 2014/15 Problem 3** ⋆⋆⋆
   Shooting lemma at the point of tangency, then the incentre-excentre lemma in triangle AQP.

5. **BMO2 2020/21 Problem 3** ⋆⋆⋆
   K is the centre of a spiral similarity taking the incircle contact points on AB and AC to B and C.

6. **BMO2 2023/24 Problem 3** ⋆⋆⋆
   The line through the two midpoints is the radical axis of the circumcircle and the point circle at P.

7. **BMO2 2024/25 Problem 2** ⋆⋆⋆
   Both circles pass through I, and so do lines BP and CQ; this is the set-up of Section 1.

8. **IMO 2005 Problem 5** ⋆⋆⋆
   Equal lengths BC = DA and BE = DF suggest a rotation; find the segments it maps onto each other.

9. **USAMO 2013 Problem 1** ⋆⋆⋆
   The three circles meet at the Miquel point; look at the spiral similarities centred there.

10. **USAMO 2006 Problem 6** ⋆⋆⋆
    Consider the Miquel point of lines AB, BC, CD, DA and what the ratio condition does to it.

11. **USAMO 2019 Problem 2** ⋆⋆⋆⋆
    Locate P through the products AP times AB and BP times AB; a symmedian of triangle AEB appears.

12. **IMO 2004 Problem 5** ⋆⋆⋆⋆
    Read both angle conditions in triangle BCD, at the vertices B and D; Section 4 is relevant.

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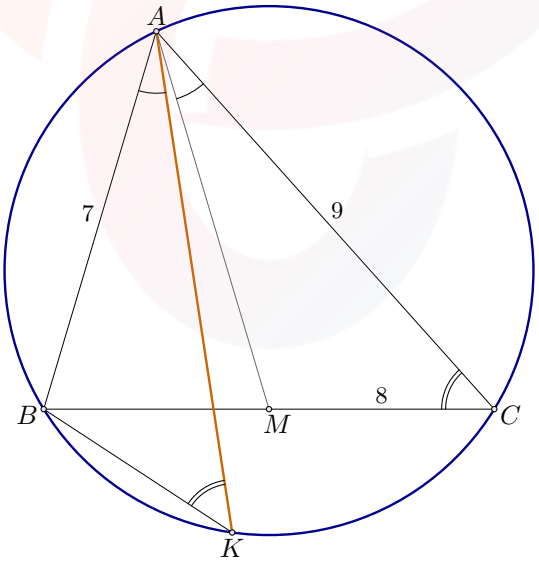
4 Hints

1. Find two similar triangles, one containing the median AM and one containing AK.
2. The spiral similarity at Q from Corollary 1.4 sends O_1 somewhere. Then pair the points differently.
3. What does the angle condition say about the isogonal conjugate of P?
4. Construct the antiparallel to BC through T, and show that there is only one segment with endpoints on lines AB, AC and midpoint T.
5. Show that M is the Miquel point. It is the centre of two spiral similarities, and the ratio of each can be read off from a pair of opposite sides. Compute MC/MA by going through MB.
6. Use the shooting lemma at B and at C, and the two circles $BKIT$, $CLIT$ of Theorem 5.2. Work with sines so that no configuration issues arise.

5 Solutions

Problem 1

Triangle ABC has $BC = 8$, $CA = 9$ and $AB = 7$. The A-symmedian meets the circumcircle again at K. Find AK.



Let M be the midpoint of BC. Since AK and AM are isogonal, $\angle BAK = \angle MAC$. Also $\angle AKB = \angle ACB = \angle ACM$, as K and C lie on the same side of AB (K is on arc BC not containing A). So triangles ABK and AMC are similar, and $AK/AC = AB/AM$, that is,

$$AK = \frac{AB \cdot AC}{AM}.$$

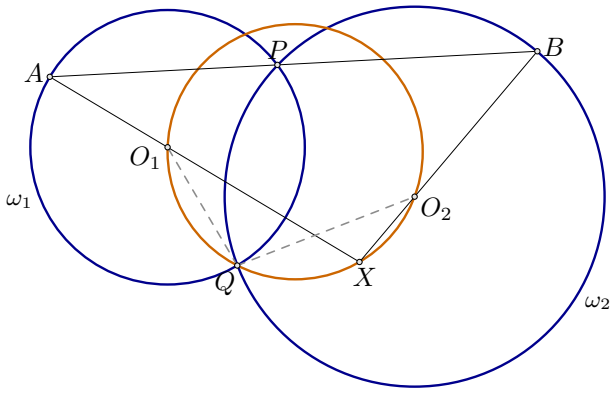
By the median formula, $AM^2 = \frac{2\cdot 9^2 + 2\cdot 7^2 - 8^2}{4} = \frac{196}{4} = 49$, so $AM = 7$ and $AK = \frac{7\cdot 9}{7} = 9$.

Answer: 9

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Problem 2

Circles ω_1 and ω_2, with centres O_1 and O_2, meet at P and Q. A line through P meets ω_1 again at A and ω_2 again at B. Prove that lines AO_1 and BO_2 meet at a point of the circle through Q, O_1 and O_2.

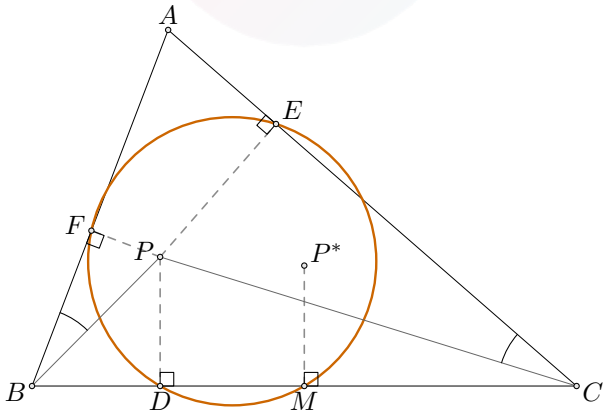


By Corollary 1.4, the spiral similarity σ with centre Q sending ω_1 to ω_2 sends A to B. A similarity sends the centre of a circle to the centre of its image, so σ sends O_1 to O_2. By Theorem 1.3, Q is also the centre of the spiral similarity sending $O_1 \mapsto A$ and $O_2 \mapsto B$.

The angle of σ is $\measuredangle O_1QO_2$, which is not 0 since Q is not on the line of centres. As σ sends line O_1A to line O_2B, these lines make the angle $\measuredangle O_1QO_2 \neq 0$ with each other and so meet at a point X. Now apply Theorem 1.2 to the similarity $O_1 \mapsto A$, $O_2 \mapsto B$: the lines joining corresponding points are O_1A and O_2B, meeting at X, so the centre Q lies on circle XO_1O_2. Hence X lies on circle QO_1O_2. (If $X = O_1$ or $X = O_2$ the conclusion holds trivially.)

Problem 3

Point P lies inside triangle ABC with $\angle PBA = \angle PCA$. Let D, E, F be the feet of the perpendiculars from P to BC, CA, AB, and let M be the midpoint of BC. Prove that D, E, F, M lie on a circle.



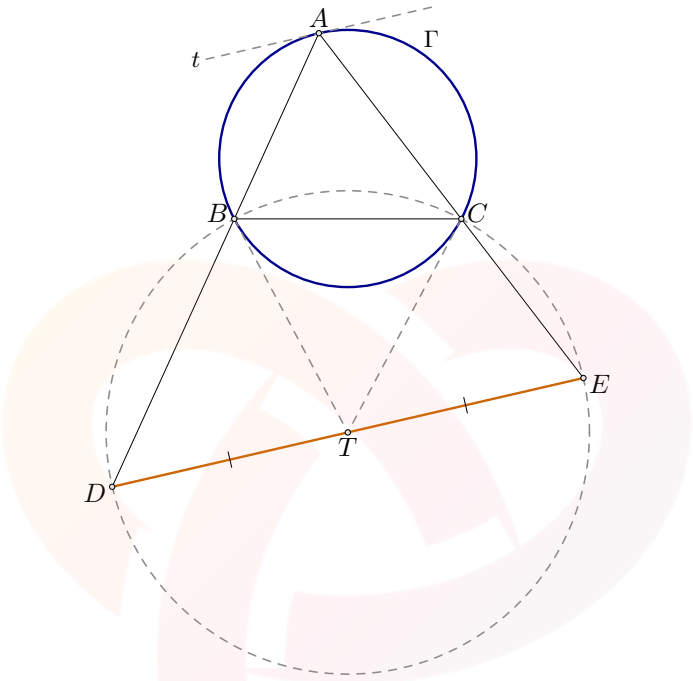
Let P^* be the isogonal conjugate of P; it exists and lies inside the triangle by Theorem 4.1, since each reflected line lies inside the corresponding angle. By definition $\angle P^*BC = \angle PBA$ and $\angle P^*CB = \angle PCA$. The hypothesis therefore gives $\angle P^*BC = \angle P^*CB$, so triangle P^*BC is isosceles with $P^*B = P^*C$, and P^* lies on the perpendicular bisector of BC. Hence the foot of the perpendicular from P^* to BC is M.

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By Theorem 4.1 the feet of the perpendiculars from P and from P^* to the three sides lie on one circle. In particular D, E, F (feet from P) and M (a foot from P^*) are concyclic. If $M = D$ the statement is about three points and is trivial.

Problem 4

Let ABC be an acute triangle with circumcircle Γ. The tangents to Γ at B and C meet at T. Points D on line AB and E on line AC are such that T is the midpoint of DE. Prove that B, C, E, D are concyclic.



Since $\angle A \neq 90^\circ$, T exists, and by Theorem 3.1(3) line AT is the A-symmedian.

Let t be the tangent to Γ at A and let m be the line through T parallel to t. The line t is not parallel to AB or AC, so m meets AB at a point D_1 and AC at a point E_1. Also $m \neq t$, because T is not on t (from T there are only the two tangents TB and TC), so $D_1, E_1 \neq A$. Further $D_1 \neq B$: otherwise $m = TB$ would be the tangent at B and would be parallel to t, so AB would be a diameter and $\angle C = 90^\circ$, contrary to the triangle being acute. Similarly $E_1 \neq C$.

By the alternate segment theorem $\measuredangle(t, AB) = \measuredangle ACB$, so $\measuredangle(D_1E_1, D_1B) = \measuredangle(t, AB) = \measuredangle ACB = \measuredangle(E_1C, CB)$, which is the directed-angle criterion for B, C, E_1, D_1 to be concyclic. So D_1E_1 is an antiparallel to BC, and by Theorem 3.1(4) the symmedian AT bisects it; since T lies on D_1E_1, T is its midpoint.

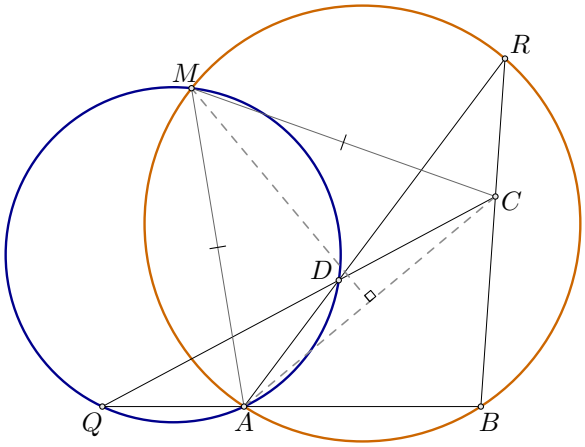
It remains to show $D = D_1$ and $E = E_1$. If T is the midpoint of DE, then E is the reflection of D in T, so E lies on the reflection of line AB in T. That reflection is a line parallel to AB, so it meets line AC in exactly one point. Hence E, and then D, are determined uniquely by the condition, and $(D, E) = (D_1, E_1)$. Therefore B, C, E, D are concyclic.

Problem 5

Let $ABCD$ be a convex quadrilateral in which no two sides are parallel. Lines AB and CD meet at Q, and lines AD and BC meet at R. Prove that the circumcircles of triangles QAD and RAB meet at a second point M, and that M lies on the perpendicular bisector of AC if and only if

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$$AB \cdot AD = CB \cdot CD.$$



First the configuration. No three of A, B, C, D are collinear, so Q is not on line AD (else $Q = A$, and A would lie on line CD), and the triangle QAD exists; similarly triangle RAB exists. By Theorem 2.1 the circles QAD, QBC, RAB, RCD pass through the Miquel point M_0 of $ABCD$. Note that $M_0 \neq A$: circle RCD meets line AD only at R and D, and $A \neq R$ because R lies on line BC. The circles QAD and RAB are different, since circle RAB meets line AD only at R and A and so does not contain D. Hence they meet at exactly the two points A and M_0, and $M = M_0$.

We also need M to differ from each vertex. Circle QAD meets line AB only at Q, A and line CD only at Q, D, and $B, C \neq Q$; so $M \neq B$ and $M \neq C$. Circle RAB meets line AD only at R, A, and $D \neq R$; so $M \neq D$. With $M \neq A$ from above, the four distances MA, MB, MC, MD are positive.

Now the ratios. By Theorem 2.1, M is the centre of the spiral similarity σ with $A \mapsto B$, $D \mapsto C$, and of the spiral similarity τ with $A \mapsto D$, $B \mapsto C$. A spiral similarity with centre M and ratio k multiplies every distance by k, distances from M included. The map σ sends segment AD to segment BC and A to B, so its ratio is

$$\frac{BC}{AD} = \frac{MB}{MA}.$$

The map τ sends segment AB to segment DC and B to C, so its ratio is

$$\frac{DC}{AB} = \frac{MC}{MB}.$$

Multiplying the two equations gives, for every quadrilateral in the problem,

$$\frac{MC}{MA} = \frac{CB \cdot CD}{AB \cdot AD}.$$

Finally, M lies on the perpendicular bisector of AC if and only if $MA = MC$, that is, if and only if the left side above equals 1, and this happens if and only if $AB \cdot AD = CB \cdot CD$. Both directions of the equivalence come from the one identity.

The same two ratios give $MB/MD = (BA \cdot BC)/(DA \cdot DC)$ and $MA \cdot MC = MB \cdot MD$. For a cyclic quadrilateral, $\sin \angle BAD = \sin \angle BCD$, so the condition says that triangles ABD and CBD have equal area, that is, BD passes through the midpoint of AC. This agrees with Section 2: there M is the inverse of $AC \cap BD$ in the circumcircle, so M lies on the line through the centre and the midpoint of AC, which is the perpendicular bisector of AC.

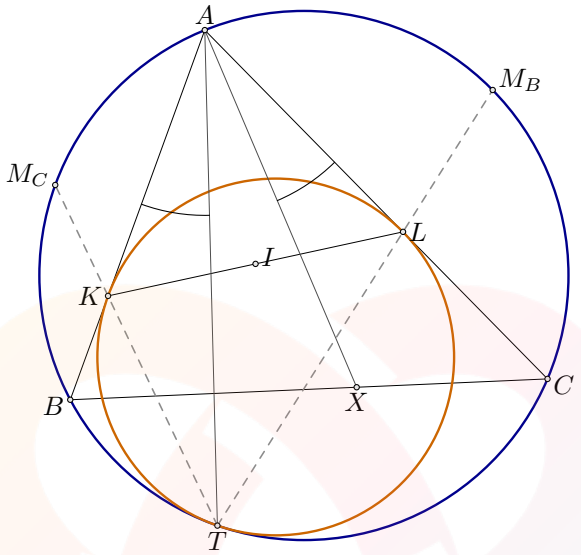
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Problem 6

The A-mixtilinear incircle of triangle ABC touches the circumcircle at T, and the A-excircle touches BC at X. With $a = BC$, $b = CA$, $c = AB$ and s the semiperimeter, prove that

$$\frac{TB}{TC} = \frac{c(s-b)}{b(s-c)},$$

and deduce that $\angle BAT = \angle XAC$.



Use the notation of Section 5: K, L are the points of contact on AB, AC, and R is the circumradius.

Step 1. By the shooting lemma, T, K, M_C are collinear and $M_CK \cdot M_CT = M_CB^2$, so triangles M_CBK and M_CTB are similar and $\frac{BK}{TB} = \frac{M_CB}{M_CT}$. Likewise $\frac{CL}{TC} = \frac{M_BC}{M_BT}$. Hence

$$\frac{TB}{TC} = \frac{BK}{CL} \cdot \frac{M_CT}{M_BT} \cdot \frac{M_BC}{M_CB}.$$

Step 2. The chords M_BC and M_CB subtend the angles $\frac{B}{2}$ and $\frac{C}{2}$ at B and C, so $\frac{M_BC}{M_CB} = \frac{\sin(B/2)}{\sin(C/2)}$. The chord M_CT subtends $\angle M_CCT = \angle ICT$ at C (as I lies on segment CM_C), and M_BT subtends $\angle IBT$ at B. In circle $CLIT$, $\sin\angle ICT = \sin\angle ILT$, and in circle $BKIT$, $\sin\angle IBT = \sin\angle IKT$ (inscribed angles on the same chord are equal or supplementary). As I lies on segment KL, the sine rule in triangle TKL gives

$$\frac{M_CT}{M_BT} = \frac{\sin\angle ILT}{\sin\angle IKT} = \frac{\sin\angle KLT}{\sin\angle LKT} = \frac{TK}{TL}.$$

Step 3. Let ρ_1, ρ_2 be the radii of circles $BKIT$ and $CLIT$. Then $TK = 2\rho_1 \sin\angle TBA$ and $TL = 2\rho_2 \sin\angle TCA$, and these sines are equal because $ABTC$ is cyclic. So $\frac{TK}{TL} = \frac{\rho_1}{\rho_2}$. In the proof of Theorem 5.2 we found $\angle KIB = \frac{C}{2}$, and similarly $\angle LIC = \frac{B}{2}$, so $\rho_1 = \frac{BK}{2\sin(C/2)}$, $\rho_2 = \frac{CL}{2\sin(B/2)}$ and

$$\frac{TK}{TL} = \frac{BK}{CL} \cdot \frac{\sin(B/2)}{\sin(C/2)}.$$

Step 4. Combining, $\frac{TB}{TC} = \left(\frac{BK}{CL}\right)^2 \frac{\sin^2(B/2)}{\sin^2(C/2)}$. From Theorem 5.2, $AK = AL = bc/s$, so $BK = \frac{c(s-b)}{s}$ and $CL = \frac{b(s-c)}{s}$. The half-angle formulas give $\sin^2\frac{B}{2} = \frac{(s-a)(s-c)}{ac}$ and $\sin^2\frac{C}{2} = \frac{(s-a)(s-b)}{ab}$. Therefore

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$$\frac{TB}{TC} = \frac{c^2(s-b)^2}{b^2(s-c)^2} \cdot \frac{b(s-c)}{c(s-b)} = \frac{c(s-b)}{b(s-c)}.$$

Step 5. Let $x = \angle BAT$, so $\angle TAC = A - x$, as T lies on arc BC not containing A. By the extended sine rule $\frac{\sin x}{\sin(A-x)} = \frac{TB}{TC}$. Let $y = \angle XAC$. Since $BX = s - c$ and $XC = s - b$, the ratio lemma gives $\frac{s-c}{s-b} = \frac{c\sin(A-y)}{b\sin y}$, so $\frac{\sin y}{\sin(A-y)} = \frac{c(s-b)}{b(s-c)} = \frac{\sin x}{\sin(A-x)}$. The function $t \mapsto \frac{\sin t}{\sin(A-t)}$ is increasing on $(0, A)$, since its derivative is $\frac{\sin A}{\sin^2(A-t)} > 0$. Hence $x = y$, that is, $\angle BAT = \angle XAC$.